United States Patent	12409961
Kind Code	B2
Date of Patent	September 09, 2025
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Tray insertion system and method

Abstract

An apparatus for packing a plurality of filled produce trays into a transport container, such as a box or a carton, without bruising the produce in the process. The produce packing system includes an infeed belt, produce information sensors and a processing assembly. The processing assembly receives information from the produce information sensors to calculate a container packing position. Using this information, the produce packing system clamps onto a filled produce tray from an infeed belt and moves the filled produce tray downward into a transport container at a predetermined height to place the filled produce tray into the transport container.

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Appl. No.: **18/504145**

Filed: **November 07, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240150049 A1	May. 09, 2024

Related U.S. Application Data

us-provisional-application US 63382958 20221109

Publication Classification

Int. Cl.: **B65B57/20** (20060101); **B65B5/06** (20060101); **B65B5/10** (20060101); B65B25/04 (20060101)

U.S. Cl.:

CPC **B65B5/068** (20130101); **B65B5/108** (20130101); **B65B57/20** (20130101); B65B25/046 (20130101)

Field of Classification Search

USPC: 53/443

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application claims priority from the following U.S. patent applications: this application claims priority to, and the benefit of, U.S. Provisional Patent Application Ser. No. 63/382,958, filed Nov. 9, 2022, which is incorporated by reference herein in its entirety.

BACKGROUND
(1) Produce, once harvested, is typically taken to a warehouse where it is sorted, sized and packaged for transportation and storage purposes. Certain types of produce, when packaged, use or require a tray insert to properly orient the produce in a box or crate. Apples and stone fruits are examples of produce typically packaged this way.
(2) The trays used in the packaging process are typically thin molded profiles with cups for each piece of fruit to rest in. Depending on the size of the produce, the cup pattern and tray size will differ. The packaging process requires each tray to be placed inside of the box or crate once every cup on the tray has been filled. This process repeats until the appropriate number of trays are loaded into the box. The number of trays varies with the size of the produce and box or crate used.
(3) This box filling operation has generally been done by hand; making it very labor-intensive, slow and expensive. Further, filling the boxes this way creates an environment where workers are more prone to repetitive stress injuries. Efforts have been made to automate this process, but several hurdles make it difficult. The trays and boxes commonly used are defined by industry standards or practice, and often, the trays used in the industry are wider than the opening of the box they are being put into. This makes it difficult to automate the process because the tray does not fit easily in the box. The trays are extremely pliable and prone to tearing when holding the weight of

the produce. Also, the produce being handled in this type of packing operation is very susceptible to bruising. The tray being inserted on top of a tray already in the box cannot be dropped on the tray below it. To do so would certainly damage the produce on the underlying tray. Because of these limitations, prior attempts to automate the packing process for these types of produce have had significant drawbacks.

(4) Accordingly, there is a need for a produce packing system and method that automates packing filled produce trays in boxes without bruising the produce in the process.

SUMMARY

(5) According to one aspect of the present invention, an apparatus for packing a number of filled produce trays into a transport container, includes a produce information sensor; a processing assembly, in communication with the produce information sensor, configured to receive information from the produce information sensor to calculate a container packing position; a linear actuator assembly; and an end effector attached to an end of the linear actuator assembly. According to this aspect of the present invention, the processing assembly activates the end effector to clamp onto a filled produce tray provided to the end effector; activates the linear actuator assembly to move the filled produce tray downward into a transport container and, at the container packing position, unclamps the end effector from the filled produce tray to place the filled produce tray into the transport container.

(6) According to another aspect of the present invention, the apparatus for packing a number of filled produce trays into a transport container may have several produce information sensors and the produce information sensors may scan for height parameters of a filled produce tray and save the height parameter data to the processing assembly. The processing assembly may calculate the container packing position from the saved height parameter data. According to yet another aspect of the present invention, the apparatus for packing a number of filled produce trays into a transport container may have a retractable infeed belt and in addition may have several tray location sensors.

(7) According to another aspect of the present invention, the apparatus for packing a number of filled produce trays into a transport container may have several justifiers, and the justifiers may be integrated with the produce information sensor into a common assembly. The end effector of the apparatus may also include a vacuum mechanism for engaging a filled produce tray.

(8) According to yet another aspect of the present invention, a method for packing a number of filled produce trays into a transport container, includes providing a produce packing system; having a retractable infeed belt; several produce information sensors; a processing assembly, in communication with the produce information sensors, configured to receive information from the produce information sensors; a linear actuator assembly; and an end effector attached to an end of the linear actuator assembly having a vacuum mechanism; and providing one filled produce tray; wherein the produce information sensors, scan the one filled produce tray to gather data about the one filled produce tray; calculate a container packing position from the gathered data about the one filled produce tray; activate the end effector to clamp onto and provide suction to the one filled produce tray; activate the linear actuator assembly to move the one filled produce tray downward into a transport container; and deactivate the suction and unclamp the end effector from the one filled produce tray at the calculated container packing position to place the one filled produce tray into the transport container.

(9) According to another aspect of the present invention, the method for packing a number of filled produce trays into a transport container, where the one filled produce tray is one of a number of filled produce trays; the processing assembly calculates a container packing position for each of the number of filled produce trays and places each of the number of filled produce trays at the respective calculated container packing position in the transport container; the processing assembly stores a maximum amount of filled produce trays that can be stored in the transport container; and the processing assembly tracks the amount of filled produce trays that have been placed in the transport container; where when the processing assembly determines that the stored maximum

amount of filled produce trays that can be stored in the transport container is recached, the processing assembly stops the process for that transport container, such that no more filled produce trays can be placed in the transport container. According to another aspect of the present invention, the method for packing a number of filled produce trays into a transport container, where the produce packing system further includes a number of justifiers that are integrated with the number of produce information sensors in a common assembly; and aligns the one filled produce tray in preparation for the end effector while the produce packing system scans the one filled produce tray to gather data about the next one filled produce tray.

Description

DRAWINGS
(1) Objects, features, and advantages of the present invention will become apparent upon reading the following description in conjunction with the drawing figures, in which:
(2) FIG. **1** is a perspective view of an embodiment of a produce packing system of the present invention;
(3) FIG. **2** is an end view of an embodiment of a produce packing system of the present invention;
(4) FIG. **3** is a cross sectional view along cross section **3-3** of FIG. **2**;
(5) FIG. **4** is a block diagram illustrating a configuration of an exemplary control system of a produce packing system of the present invention;
(6) FIG. **5** is a perspective view of an end effector of an embodiment of a produce packing system of the present invention;
(7) FIG. **6** is a side view of an end effector of an embodiment of a produce packing system of the present invention;
(8) FIG. **7** is an end view of an end effector of an embodiment of a produce packing system of the present invention;
(9) FIG. **8** is a flowchart illustrating the flow of an embodiment of a process executed by the produce packing system of the present invention;
(10) FIG. **9** is a side sectional view depicting a step in the process of the produce packing system of the present invention;
(11) FIG. **10** is a flowchart illustrating the flow of an embodiment of a container packing parameter position calculation process of the produce packing system of the present invention;
(12) FIG. **11** is a descriptive view depicting sensor information collection in a container packing parameter position calculation process of the produce packing system of the present invention;
(13) FIG. **12** is a top plan view depicting an alignment step in the process of the produce packing system of the present invention;
(14) FIG. **13** is a top plan view depicting another alignment step in the process of the produce packing system of the present invention;
(15) FIG. **14** is a side sectional view depicting a step in the process of the produce packing system of the present invention;
(16) FIG. **15** is a side sectional view depicting a step in the process of the produce packing system of the present invention;
(17) FIG. **16** is a side sectional view depicting a step in the process of the produce packing system of the present invention;
(18) FIG. **17** is a side sectional view depicting an end effector of the present invention clamping a filled produce tray;
(19) FIG. **18** is a side sectional view depicting a step in the process of the produce packing system of the present invention;
(20) FIG. **19** is a side sectional view depicting a step in the process of the produce packing system

of the present invention;
(21) FIG. **20** is a flowchart illustrating the flow of an embodiment of a tray placement process of the produce packing system of the present invention;
(22) FIG. **21** is a side sectional view depicting a step in the process of the produce packing system of the present invention;
(23) FIG. **22** is a side sectional view depicting a step in the process of the produce packing system of the present invention;
(24) FIG. **23** is a side sectional view of another embodiment of the produce packing system of the present invention;
(25) FIG. **24** is a top plan view depicting an alignment step in the process of another embodiment of the produce packing system of the present invention; and
(26) FIG. **25** is a top plan view depicting another alignment step in the process of another embodiment of the produce packing system of the present invention.
DESCRIPTION
(27) Referring to FIGS. **1**, **2** and **3**, an embodiment of a produce packing system **10** of the present invention is depicted. Referring specifically to FIG. **3**, the produce packing system **10** depicted includes, among other items, an infeed belt **20**, an end effector **22**, a machine control box **23**, a linear actuator assembly **24**, produce information sensors **26***a*-**26***c* (FIGS. **3**, **12**), location sensors **28***a*-**28***c*, an extraction belt **30**, a vacuum pump **27**, justifiers **32***a*-**23***c* (FIGS. **3**, **13**) and a processing assembly **34** contained within the machine control box **23**. In this embodiment, the other half of the produce packing system **10** not depicted in FIG. **3** has similar components to the portion depicted. As explained in detail below, the locations sensors **28***a*-**28***c* locate a tray **50** (FIG. **9**) of produce and, with the justifiers **32***a*-**32***c*, control the infeed belt **20**, starting and stopping the belt **20** at desired times and positions. It should be understood that the number and type of produce information sensors **26** and location sensors **28** may vary depending on application and need and there are numerous various embodiments of the present invention using different sensor configurations. Referring to FIG. **4**, in this embodiment, the processing assembly **34** includes a control unit **35**, a processor **36** and a data storage unit **38**. Further, in this embodiment, in addition to other items, the produce information sensors **26***a*-**26***c* and the location sensors **28***a*-**28***c* communicate with the processing assembly **34**.
(28) As explained in detail below, the end effector **22** is the mechanism that engages a tray **50** of produce and moves it into a transport container, such as a box or crate. The end effector **22** has a number of components. Referring to FIGS. **5**, **6** and **7**, an embodiment of the end effector **22** of the present invention includes actuators **40**, major clamp plates **44**, minor clamp plates **46** and a vacuum mechanism **60**. Each minor clamp plate and major clamp plate **44**, **46**, in this embodiment, has slots **48** formed in their sidewalls to allow each clamp plate **44**, **46** to move freely upward and downward until engaged. It should be understood that in other embodiments the position, style, and number of pneumatic actuators **40** could be changed. Similarly, in other embodiments, the look, flange angles, lengths and adjustability of the minor and major clamp plates **44**, **46** may vary as well.
(29) Referring now to FIGS. **8-22**, the operation of an embodiment of a produce packing system **10** of the present invention is explained. FIG. **8** depicts a flowchart of an embodiment of the process implemented by the control unit **35** during a cycle of the produce packing system **10**. When turned on, the produce packing system **10** remains in a standby state until activated by a user. A user activates the produce packing system **10** by entering the desired number of trays per box and pressing the start button on an LCD screen. The information entered by the user is stored in data storage unit **38** and used by the control unit **35** to determine the parameters of the cycle. Once the user enters the cycle information and places a tray of produce **50**, in this example, apples, on the infeed belt **20** of the produce packing system **10**, the user presses a start button and the machine starts the process shown in FIG. **8** and in the position shown in FIG. **9**.

(30) The cycle is started (step **100**). At step **102**, the system is checked for faults. If a fault is detected, the process alerts the user and prompts the user to check for faults (steps **104**, **106**). If at step **102**, no faults are detected, the process, at step **108**, sets all systems to zero, homes the actuator components to home state and all functions are enabled. Then, at step **110**, if the sensor **28***a* does not detect a tray **50**, the control unit **35** signals the infeed belt **20** to advance (step **112**). However, if at step **110**, the sensor **28***a* does detect a tray **50**, the process, at step **114**, stops the infeed belt **20**.
(31) The tray **50**, if the first tray, is now ready to begin the first placement cycle. If the tray **50** is one or more of the follow-on trays, the cycle continues until complete as set by the user's parameters. At step **116**, the process checks the linear actuator **24** to determine if it is in the fully retracted home position. If not, at step **118**, the process checks if the major clamps **44** and minor clamps **46** located on the end effector **22** are extended. If the clamps **44** and **46** are extended, at step **120**, the process retracts the linear actuator **24**. If the clamps **44** and **46** are not extended, the process proceeds to step **114** again. The process continues in this loop until the linear actuator **24** is in the home position as shown in FIG. **9**. With the linear actuator **24** in the home position, the process, at step **122**, re-starts advancement of the retractable belt **20**. Then, at step **124**, when the produce information sensors **26***a*-**26***c* detect a tray **50** (FIG. **11**), the process executes the product height detection process.
(32) Referring now to FIGS. **10** and **11**, the process starts the product height detection process. At steps **126**, the belt **20** is extending outward, with the tray **50** on it. The tray **50** travels underneath the produce information sensors **26***a*-**26***c* (e.g., height sensors in this embodiment) and the height sensors **26***a*-**26***c* each transmit the information detected to the processing assembly **34**. As the tray **50** passes under the height sensors **26***a*-**26***c*, the process, at step **128**, using the sensors **26***a*-**26***c*, measures the height of the produce on the tray **50**. The process, using the information from the sensors **26***a*-**26***c*, repetitively compares the distance from each sensor **26** to the retractable belt **20** (FIG. **11**, "H**1**"), to the dynamic heights (FIG. **11**, "H**2**" and "H**3**") from each sensor **26**. In the embodiment depicted in FIG. **11**, at step **130**, the process transmits the difference between H**2** and H**1** to the data storage unit **38**. The difference between H**3** and H**1** is overridden since it is a smaller value than the difference between H**2** and H**1**. The process continues to measure the difference between H**2** and H**1** as the tray **50** moves under the sensors **26***a*-**26***c*. A difference greater than the previous one is constantly being evaluated and stored until the location sensor **28***b* detects the tray **50**, (step **132**). The process continues to overwrite the previous H**2** minus H**1** value if the previous H**2**/H**1** value is smaller than the newly measured H**2**/H**1** value. The process ultimately calculates the largest H**2** minus H**1** value for the tray **50** under the sensors **26***a*-**26***c* for use in calculating where the tray **50** should be placed in the box **64**. The process, then, at step **134**, deactivates produce information sensors **26***a*-**26***c* and the product height detection process ends. It should be understood that the information gathered by sensors **26** can be any information that a user wants to collect. In this embodiment, the produce height sensors **26***a*-**26***c* gather information that the processing assembly **34** uses to determine the size or height of the apples on the tray **50** as previously described.
(33) Referring again to FIG. **8**, the retractable belt **20** has been continuing to advance through the product height detection process and continues to advance forward towards the end effector **22** until both sensors **28***b*, **28***c* detect the tray **50** at step **136** (FIG. **12**), at which point, the process stops the retractable belt **20** (step **138**). Referring now to FIGS. **12** and **13**, the process, at step **140**, cycles the tray justifiers **32***a*, **32***b* and **32***c* to align the tray **50** underneath the linear actuator **24**. The process, at step **140**, ensures that any misalignment of the tray **50** is corrected before engaging the tray **50** with the linear actuator **24**. Referring to FIG. **14**, the tray **50** is now aligned with the overhead linear actuator **24** and ready to be placed in a box **64** below.
(34) At step **142**, the process extends the linear actuator **24** downward to the stored average height value determined by the processing assembly at step **124** (FIG. **15**). At this point, at step **144**, the process activates the vacuum pump **27** which creates suction on the tray **50** through the vacuum

mechanism **60** (FIG. **17**). Next, at step **146**, the process retracts the major **44** and minor **46** side clamps **44**, **46** of the end effector **22** ("EOT") (FIG. **16**). Referring now to FIGS. **7** and **17**, in order to account for different sized products, the minor and major clamp plates **44**, **46** can freely slide up and down along slots **48** when the pneumatic actuators **40** are in the extended position. This configuration allows the vacuum mechanism **60** of the end defector **22** to reach and form to the produce, apples in this embodiment, at several heights depending on their size, with automatic adjustment of the side clamps **44**, **46**. The side clamps **44**, **46** then retract, and the vacuum mechanism **60** engages the tray **50**. This process of clamping the tray **50** at the optimal height while applying a vacuum, holds the tray **50** in a relatively flat or horizontal position, as depicted in FIG. **17**. With the produce packaging system **10** of the present invention, as explained herein, the three aspects of adjustability, grip pressure and vacuum working together keeps the tray **50** from sagging in the middle while engaged by the end effector **22**, which allows the tray **50** to be placed in the box **64**.

(35) Referring now to FIGS. **8** and **18**, at step **148**, the linear actuator **24** now retracts to a preset height. At step **150**, the process checks to see if the tray **50** has been picked up using feedback from sensors **28***b* and **28***c*. If the tray **50** is detected by the sensors **28***b* or **28***c*, the process faults, reverting back to step **102**. If sensors **28***b* and **28***c* do not detect the tray **50**, then, at step **152** (FIG. **19**), the process retracts the retractable belt **20**. Then, at step **154**, the process executes the tray placement process.

(36) Referring now to FIG. **20**, at step **155**, the process starts the tray placement process. At step **156**, the process checks to see how many trays **50** have been placed in the box **64**. It is important to note that, in this embodiment, at this point in the process, the process simultaneously starts again at step **100** for a new tray **50** in order to have the next tray **50** ready when the linear actuator **24** returns home. At step **156**, if the number of trays **50** matches the required trays **50** per box **64**, the process ends at step **179** because the tray packing cycle is complete. If the number of trays **50** in the box is less than the desired trays in the box, at step **164**, the current height of tray **50** in end effector **22** is gathered from the storage unit **38**. At step **166**, the process retrieves that height value and compounds it with the previously stored compounded height values. For example, if there are five required trays **50** per box **64** and three have been placed so far with the fourth tray **50** being placed. The stored height values from each of the last three trays along with the height value of the current tray **50** in the end effector **22** are all compounded and stored. All tray height values are the tray heights found and stored during the product height detection process at step **124**. Whether the desired trays **50** per box **64** was met or not, at step **168**, the process, in this embodiment, takes the updated tray height either compounded or first tray **50** and subtracts that value from the presently known distance between the end effector **22** and the top of the box conveyor belt **30** to calculate a container packing position. Then, at step **170**, the process extends the linear actuator **24** to that new calculated position (FIG. **21** for this example; showing the first tray **50** being placed in the box **64**). Once the linear actuator **24** is at the precisely calculated height, the process, at step **172**, extends the major and minor side clamps **44** and **46**, releasing the tray **50** as shown in FIG. **22**. The process at step **174** disengages the vacuum pump **27**, in turn releasing the suction in the vacuum mechanism **60**. The linear actuator **24** then is retracted in step **176**, returning the produce packing system **10** to the position depicted in FIG. **9**. At step **178**, a unit of one is added to the number of trays in the box count, and at step **179**, the tray placement process ends. The process reverts back to step **110** in FIG. **8** where the next tray **50**, in this embodiment, is already in process.

(37) It should be noted that keeping the tray **50** from sagging using a vacuum, as explained above, allows the linear actuator **24** to lower the tray to the desired position without rubbing, bruising, smashing the apples on the tray **50** in the end effector **22** or the tray **50** below it in the box **64**. It should be noted that typically the trays **50** used are normally ¼ to ½ in larger diameter than the inner dimensions of the box, making it difficult to properly place a tray **50** in the box **64** without bruising the produce. The produce packing system **10** of this invention overcomes this by

accounting for this size differential when disengaging the end effector **22** to place the tray in the box **64**.

(38) Referring now to FIGS. **23-25**, another embodiment of the produce packing system **10** of the present invention is depicted. In this embodiment, instead of the process of product height detection (FIGS. **8**, **10**) and the tray adjustment (FIGS. **8**, **12-13**) happening at different stages of the advancement of the infeed belt **20**, this embodiment includes an integrated unit **70** that integrates the produce information sensors **26***a*-**26***c* and the justifiers **32***b*, **32***c* so that the produce packing system **10** can detect product height and align the tray **50** at the same time, with the infeed belt **20** in the retracted position. As such, in this embodiment, the tray **50** is aligned before being situated under the linear actuator assembly **24** and the end effector **22**. This allows the tray **50** to be prepped and justified while the linear actuator assembly **24** is completing the cycle of placing the prior tray **50** into the box **64**. By integrating the process of product height detection and the tray adjustment, the cycle time between the picking and placing of trays **50** in the box **64** is decreased, leading to faster operation of the produce packing system **10**.

(39) Although the process has been described based on the embodiments disclosed and explained above, it should be noted that this process may be altered without escaping the intended scope of the described process. A few examples are, the measurement sensor or sensors **26** may be accompanied by or replaced by a bar code scanner, infrared, laser, sonar or other technology to achieve the same distance or size sensing functionality. The actuators **44** may have more degrees of freedom than just vertical movement. The infeed belt **20** may be retractable or stationary. The justifiers **32** may be a different mechanical design, but still have the same extend and retract function to square the tray **50** under the end effector **22**.

Claims

1. An apparatus for packing a plurality of open-top trays into a transport container, wherein each open-top tray contains individual pieces of produce, comprising: at least one produce information sensor for measuring the height of each individual piece of produce for each open-top tray that passes under the at least one produce information sensor; a processing assembly, in communication with the at least one produce information sensor, configured to receive measured height information from the at least one produce information sensor for each open-top tray that passes under the at least one produce information sensor, wherein the processing assembly uses the received measured height information for each open-top tray to calculate a container packing position height for a next open-top tray that is to be stacked on top of a presently measured open-top tray based on the highest piece of produce in the open-top tray and stores that calculated container packing position height for the processing assembly to use when stacking the next open-top tray; a linear actuator assembly; and an end effector attached to an end of the linear actuator assembly; wherein, when activated: the at least one produce information sensor measures the height of each individual piece of produce for a first open-top tray that passes under the at least one produce information sensor: the processing assembly receives the measured height information from the at least one produce information sensor for the first open-top tray; uses the received measured height information for the first open-top tray to calculate a first container packing position height for a second open-top tray that is to be stacked next on top of the first open-top tray, wherein the first calculated container packing position height for the second open-top tray is based on the measurement of the highest piece of produce in the first open-top tray; the processing assembly stores the first calculated container packing position height for the processing assembly to use when stacking the second open-top tray; the processing assembly then activates the end effector to clamp onto the first open-top tray; the processing assembly then activates the linear actuator assembly to move the first open-top tray into a transport container and unclamps the end effector from the first open-top tray to place the first open-top tray into the transport container; the

processing assembly retracts the linear actuator assembly to a pre-set position; then, when a second open-top tray is detected, the at least one produce information sensor measures the height of each individual piece of produce for the second open-top tray that passes under the at least one produce information sensor: the processing assembly receives the measured height information from the at least one produce information sensor for the second open-top tray; uses the received measured height information for the second open-top tray to calculate a second container packing position height for a third open-top tray that is to be stacked next on top of the second open-top tray, wherein the calculated container packing position height for the third open-top tray is based on the measurement of the highest piece of produce in the second open-top tray; the processing assembly stores the second calculated container packing position height for the processing assembly to use when stacking the third open-top tray; the processing assembly then activates the end effector to clamp onto the second open-top tray; the processing assembly then activates the linear actuator assembly to move the second open-top tray into the transport container; stops the linear actuator assembly in the transport container at the stored first calculated container packing position height, and unclamps the end effector from the second open-top tray to place the second open-top tray into the transport container without bruising, or at least minimizing bruising to, the individual pieces of produce in the first open-top tray already in the transport container; and the processing assembly continues to place open-top trays into the transport container using a calculated container packing position height from the open-top tray placed into the transport container just prior to the present open-top tray being placed until all of the open-top trays presented to the at least one produce information sensor are packed into the transport container.
2. The apparatus for packing a plurality of open-top trays into a transport container of claim 1, wherein the at least one produce information sensor comprises a plurality of produce information sensors.
3. The apparatus for packing a plurality of open-top trays into a transport container of claim 2, wherein the plurality of produce information sensors measure the height of each individual piece of produce for each tray that passes under the plurality of produce information sensors.
4. The apparatus for packing a plurality of open-top trays into a transport container of claim 3, wherein the processing assembly calculates a container packing position from the measured height information that the processing assembly receives from the plurality of produce information sensors.
5. The apparatus for packing a plurality of open-top trays into a transport container of claim 1, further comprising a retractable infeed belt.
6. The apparatus for packing a plurality of open-top trays into a transport container of claim 5, further comprising a plurality of tray location sensors.
7. The apparatus for packing a plurality of open-top trays into a transport container of claim 1, further comprising a plurality of justifiers.
8. The apparatus for packing a plurality of open-top trays into a transport container of claim 1, wherein the at least one produce information sensor and the plurality of justifiers are integrated in a common assembly.
9. The apparatus for packing a plurality of open-top trays into a transport container of claim 1, wherein the end effector includes a vacuum mechanism for engaging an open-top tray.
10. An apparatus for packing a plurality of open-top trays into a transport container, wherein each open-top tray contains individual pieces of produce, comprising: an infeed belt; at least one produce information sensor for measuring the height of each individual piece of produce for each open-top tray that passes under the at least one produce information sensor; a processing assembly, in communication with the at least one produce information sensor, configured to receive measured height information from the at least one produce information sensor for each open-top tray that passes under the at least one produce information sensor, wherein the processing assembly uses the received measured height information for each open-top tray to calculate a container packing

position height for a next open-top tray that is to be stacked on top of a presently measured open-top tray based on the highest piece of produce in the open-top tray and stores that calculated container packing position height for the processing assembly to use when stacking the next open-top tray; a linear actuator assembly; and an end effector attached to an end of the linear actuator assembly having a vacuum mechanism; wherein, when activated: the infeed belt feeds a first open-top tray to the at least one produce information sensor; the at least one produce information sensor measures the height of each individual piece of produce for the first open-top tray that passes under the at least one produce information sensor: the processing assembly receives the measured height information from the at least one produce information sensor for the first open-top tray; uses the received measured height information for the first open-top tray to calculate a first container packing position height for a second open-top tray that is to be stacked next on top of the first open-top tray, wherein the first calculated container packing position height for the second open-top tray is based on the measurement of the highest piece of produce in the first open-top tray; the processing assembly stores the first calculated container packing position height for the processing assembly to use when stacking the second open-top tray; the processing assembly then activates the end effector to clamp onto and provide suction to the first open-top tray fed to the end effector by the infeed belt; the processing assembly then activates the linear actuator assembly to move the first open-top tray into a transport container and unclamps the end effector from the first open-top tray and deactivates the suction to place the first open-top tray into the transport container; the processing assembly retracts the linear actuator assembly to a pre-set position; then, the infeed belt feeds a second open-top tray to the at least one produce information sensor; the at least one produce information sensor measures the height of each individual piece of produce for the second open-top tray that passes under the at least one produce information sensor: the processing assembly receives the measured height information from the at least one produce information sensor for the second open-top tray; uses the received measured height information for the second open-top tray to calculate a second container packing position height for a third tray that is to be stacked next on top of the second open-top tray, wherein the calculated container packing position height for the third open-top tray is based on the measurement of the highest piece of produce in the second open-top tray; the processing assembly stores the second calculated container packing position height for the processing assembly to use when stacking the third open-top tray; the processing assembly then activates the end effector to clamp onto and provide suction to the second open-top tray fed to the end effector by the infeed belt; the processing assembly then activates the linear actuator assembly to move the second open-top tray into the transport container; stops the linear actuator assembly in the transport container at the stored first calculated container packing position height, and deactivates the suction and unclamps the end effector from the second open-top tray to place the second open-top tray into the transport container without bruising, or at least minimizing bruising to, the individual pieces of produce in the first open-top tray already in the transport container; and the processing assembly continues to place open-top trays into the transport container using a calculated container packing position height from the tray placed into the transport container just prior to the present open-top tray being placed until all of the open-top trays presented to the at least one produce information sensor are packed into the transport container.

11. The apparatus for packing a plurality of open-top trays into a transport container of claim 10, wherein the at least one produce information sensor comprises a plurality of produce information sensors.

12. The apparatus for packing a plurality of open-top trays into a transport container of claim 11, wherein the plurality of produce information sensors measure the height of each individual piece of produce for each tray that passes under the plurality of produce information sensors.

13. The apparatus for packing a plurality of open-top trays into a transport container of claim 12, wherein the processing assembly calculates a container packing position from the measured height information that the processing assembly receives from the plurality of produce information

sensors.
14. The apparatus for packing a plurality of open-top trays into a transport container of claim 10, wherein the infeed belt is retractable.
15. The apparatus for packing a plurality of open-top trays into a transport container of claim 14, further comprising a plurality of tray location sensors.
16. The apparatus for packing a plurality of open-top trays into a transport container of claim 10, further comprising a plurality of justifiers.
17. The apparatus for packing a plurality of open-top trays into a transport container of claim 10, wherein the at least one produce information sensor and the plurality of justifiers are integrated in a common assembly.
18. An apparatus for packing a plurality of open-top trays into a transport container, wherein each open-top tray contains individual pieces of produce, comprising: an infeed belt; at least one produce information sensor for measuring the height of each individual piece of produce for each open-top tray that passes under the at least one produce information sensor; a processing assembly, in communication with the at least one produce information sensor, configured to receive measured height information from the at least one produce information sensor for each open-top tray that passes under the at least one produce information sensor, wherein the processing assembly uses the received measured height information for each open-top tray to calculate a container packing position height for a next open-top tray that is to be stacked on top of a presently measured open-top tray based on the highest piece of produce in the open-top tray and stores that calculated container packing position height for the processing assembly to use when stacking the next open-top tray; a linear actuator assembly; and an end effector attached to an end of the linear actuator assembly having a vacuum mechanism, a plurality of actuators, and a plurality of clamp plates connected to the plurality of actuators; wherein, when activated: the infeed belt feeds a first open-top tray to the at least one produce information sensor; the at least one produce information sensor measures the height of each individual piece of produce for the first open-top tray that passes under the at least one produce information sensor: the processing assembly receives the measured height information from the at least one produce information sensor for the first open-top tray; uses the received measured height information for the first open-top tray to calculate a first container packing position height for a second open-top tray that is to be stacked next on top of the first open-top tray, wherein the first calculated container packing position height for the second open-top tray is based on the measurement of the highest piece of produce in the first open-top tray; the processing assembly stores the first calculated container packing position height for the processing assembly to use when stacking the second open-top tray; the processing assembly then activates the end effector to clamp onto and provide suction to the first open-top tray fed to the end effector by the infeed belt; the processing assembly then activates the linear actuator assembly to move the first open-top tray into a transport container and unclamps the end effector from the first open-top tray and deactivates the suction to place the first open-top tray into the transport container; the processing assembly retracts the linear actuator assembly to a pre-set position; then, the infeed belt feeds a second open-top tray to the at least one produce information sensor; the at least one produce information sensor measures the height of each individual piece of produce for the second open-top tray that passes under the at least one produce information sensor: the processing assembly receives the measured height information from the at least one produce information sensor for the second open-top tray; uses the received measured height information for the second open-top tray to calculate a second container packing position height for a third tray that is to be stacked next on top of the second open-top tray, wherein the calculated container packing position height for the third open-top tray is based on the measurement of the highest piece of produce in the second open-top tray; the processing assembly stores the second calculated container packing position height for the processing assembly to use when stacking the third open-top tray; the processing assembly then activates the end effector to clamp onto and provide suction to the second open-top tray fed to the

end effector by the infeed belt; the processing assembly then activates the linear actuator assembly to move the second open-top tray into the transport container; stops the linear actuator assembly in the transport container at the stored first calculated container packing position height, and deactivates the suction and unclamps the end effector from the second open-top tray to place the second open-top tray into the transport container without bruising, or at least minimizing bruising to, the individual pieces of produce in the first open-top tray already in the transport container; and the processing assembly continues to place open-top trays into the transport container using a calculated container packing position height from the tray placed into the transport container just prior to the present open-top tray being placed until all of the open-top trays presented to the at least one produce information sensor are packed into the transport container.

19. The apparatus for packing a plurality of open-top trays into a transport container of claim 18, wherein the at least one produce information sensor comprises a plurality of produce information sensors.

20. The apparatus for packing a plurality of open-top trays into a transport container of claim 18, further comprising a plurality of tray location sensors.